

IMPLEMENTATION & DEPLOYMENT PLAN
Architectural Engineering, Phased Execution, and Operational Integration
Heavy Oil Wells – Baghewala Field, Rajasthan

Smart India Hackathon (SIH) 2026
Project Execution & Deployment Specification

September 5, 2026

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1 Executive Summary & Project Scope

1.1 Project Vision & Strategic Objectives

The Baghewala Heavy Oil Field in Rajasthan presents severe operational challenges due to high oil viscosity (18° API, up to 10,000 cP at reservoir temperature) and steep thermal decay following Cyclic Steam Stimulation (CSS). The **Well-to-Surface Digital Twin** provides a closed-loop cyber-physical framework that integrates downhole elastic wave mechanics, dynamic thermal fluid dynamics, machine learning computer vision, and autonomous Variable Frequency Drive (VFD) speed control.

This Implementation Plan establishes the 16-week execution roadmap to construct, validate, and deploy the end-to-end digital twin system for Smart India Hackathon (SIH) 2026.

1.2 Key Target Metrics & Engineering KPIs

- **Rod Floating Mitigation:** Zero compressive failure events through real-time dynamic stroke rate limiters (SPM_{safe}).
- **Thermal Efficiency Boost:** 12 – 15% increase in net oil recovery per CSS injection cycle by optimizing production duration t_{soak} .
- **Downhole Simulation Latency:** < 200 ms compute time for 1D Gibbs wave equation solver across 1,200-meter rod strings.
- **Closed-Loop Speed Tuning:** < 100 ms latency for local edge fallback speed adjustment upon detecting fluid pound or rod slack.

2 Work Breakdown Structure (WBS) & Phased Execution

The project is structured into four sequential phases executed over a 16-week timeframe.

2.1 Phase 1: Ingestion Pipeline & Physics Engine Core (Weeks 1–4)

1. **Task 1.1: Telemetry Protocols & Modbus/MQTT Bridge:** Implement telemetry ingestion engine supporting Modbus RTU/TCP and MQTT messaging protocols over 10 Hz sampling rates.
2. **Task 1.2: Signal Preprocessing Suite:** Build zero-phase 4th order Butterworth filter and Savitzky-Golay moving polynomial smoother to strip mechanical VFD noise.
3. **Task 1.3: 1D Gibbs Damped Wave Solver:** Develop C++/Numba finite-difference solver for 1D wave mechanics incorporating fluid damping $c(t)$.
4. **Task 1.4: Thermal & Viscosity Physics Engine:** Implement dynamic exponential cooling $T(t)$ and Andrade equation viscosity models calibrated against Baghewala PVT fluid samples.

2.2 Phase 2: AI Diagnostic Engine & Cloud Infrastructure (Weeks 5–8)

1. **Task 2.1: Downhole Dyno Card Image Rasterizer:** Build automatic continuous card generator converting plunger force-position arrays into 224×224 tensors.
2. **Task 2.2: ResNet-18 Diagnostic Classifier:** Train convolutional neural network to classify cards into 5 diagnostic operational states (Normal, Rod Float, Fluid Pound, Gas Lock, Unset Pump).
3. **Task 2.3: Reinforcement Learning Control Agent (PPO):** Construct Proximal Policy Optimization agent in PyTorch to compute safe VFD frequency setpoints.
4. **Task 2.4: Time-Series Storage Setup:** Deploy TimescaleDB and PostgreSQL database clusters with automated data retention policies.

2.3 Phase 3: Closed-Loop Control & Edge Hardware Integration (Weeks 9–12)

1. **Task 3.1: Industrial Edge Gateway Runtime:** Deploy ONNX Runtime and C++ wave solver onto Siemens MindConnect / Advantech edge controllers.

2. **Task 3.2: Local Watchdog & Fail-Safe Override:** Implement heartbeat monitor and local fallback rule engine for autonomous offline operation.
3. **Task 3.3: gRPC High-Throughput Service Mesh:** Build gRPC interfaces for real-time inter-service micro-communication.
4. **Task 3.4: ISA-101 SCADA UI Cockpit:** Develop React.js + Plotly WebGL control room dashboard featuring live surface/downhole card overlays.

2.4 Phase 4: Field Validation, Testing & Final SIH Deployment (Weeks 13–16)

1. **Task 4.1: Hardware-in-the-Loop (HIL) Bench Testing:** Simulate wellhead SCADA signals and benchmark VFD response under synthetic disturbance scenarios.
2. **Task 4.2: Field Telemetry Calibration:** Fine-tune thermal transfer coefficients (α) using empirical production data from Baghewala wells.
3. **Task 4.3: Cyber-Security & Compliance Audit:** Validate mTLS 1.3 encryption, RBAC permissions, and SCADA safety interlocks.
4. **Task 4.4: Final SIH 2026 Demonstration Package:** Finalize project documentation, user manuals, and physical hardware demonstration rig.

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3 System Dependencies & Task Interrelationships

Task ID	Task Name	Prerequisites	Key Deliverable
T1.1 – T1.2	Telemetry Ingestion & Pre-processing	Raw Field Sensor Protocols	Cleaned Telemetry Stream (10 Hz)
T1.3 – T1.4	1D Wave & Thermal Solvers	Preprocessed Data Stream	Downhole Dyno Cards & Viscosity $\mu(t)$
T2.1 – T2.2	Vision Card Classifier (ResNet)	Generated Downhole Cards	Anomaly Diagnosis Model (.onnx)
T2.3	PPO Closed-Loop Controller	Dyno Classifier & Physics Engine	Autonomous VFD Speed Policy
T3.1 – T3.2	Edge Deployment & Watchdog	ONNX Model & Local Physics Engine	Local Edge Control Daemon
T3.4	WebGL Control Room UI	gRPC Service Mesh	ISA-101 Dashboard Suite
T4.1 – T4.4	HIL Testing & SIH Rollout	Complete Integrated Pipeline	Production-Ready Digital Twin

Table 1: Task Dependency and Deliverable Mapping

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4 Resource Allocation & Team Responsibility Matrix

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5 Risk Assessment, FMEA & Mitigation Strategies

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6 Gantt Timeline & Milestone Schedule

1. **Milestone 1 :** Fully functional numerical 1D wave engine outputting downhole dyno cards from synthetic surface inputs.
2. **Milestone 2 :** Trained ResNet-18 classifier achieving $> 95\%$ validation accuracy on dynamic dyno card anomaly datasets.

Role / Sub-Team	Primary Focus	Key Responsibilities & Assigned Tasks
Subsurface Physics Lead	Mathematical Modeling	Gibbs wave equation discretization, viscosity thermal calibration, downhole boundary condition verification (T1.3, T1.4, T4.2).
ML & Computer Vision Lead	AI Infrastructure	ResNet-18 dyno card training, PPO control policy formulation, model optimization to ONNX format (T2.1, T2.2, T2.3).
Edge & Embedded Systems Lead	Field Control Systems	Modbus/MQTT drivers, Siemens MindConnect edge setup, watchdog safety interlocks, VFD hardware control (T1.1, T3.1, T3.2).
Full-Stack SCADA UI Developer	Interface & Data Stream	TimescaleDB setup, React/WebGL control room cockpit, gRPC micro-service architecture (T2.4, T3.3, T3.4).
QA & Safety Testing Engineer	System Integration	Hardware-in-the-Loop simulation, mTLS security validation, ISA-101 visual compliance (T4.1, T4.3, T4.4).

Table 2: Team Resource Allocation and Responsibility Matrix (RACI)

Identified Risk	Severity / Probability	Root Cause	Mitigation Strategy
Telemetry Drop / Network Disruption	High / Medium	Harsh desert environmental interference on LTE/5G	Edge gateway buffers data locally and switches to autonomous offline fallback model (T3.2).
Rod Float False Negative	Critical / Low	Viscosity parameter miscalibration during sudden fluid cooling	Dual-validation: PINN physical limits enforce hard override if compressive force < 0 lbs.
Wave Solver Numerical Drift	Medium / Low	Courant-Friedrichs-Lewy (CFL) condition violation at high SPM	Adaptive time-step adjustment (Δt) dynamically enforced in C++ wave solver loop.
Model Drift on Heavy Crude	Medium / Medium	Unseen downhole card shapes during severe gas interference	Continuous MLOps pipeline triggers retraining when diagnostic confidence drops $< 88\%$.

Table 3: Failure Mode Effects Analysis (FMEA) and Risk Mitigation Matrix

3. **Milestone 3 :** Hardware edge gateway successfully receiving SCADA inputs and executing safe speed overrides within 100 ms.
4. **Milestone 4 :** Final End-to-End System Integration, SIH 2026 Demonstration, and complete technical handover package.